

EXPERIMENT 1: V–I CHARACTERISTICS OF PN JUNCTION & ZENER DIODES, AND STUDY OF CLIPPER & CLAMPER CIRCUITS

1. Aim

- To plot the forward- and reverse-bias V–I characteristics of a PN junction (rectifier) diode and to determine its cut-in voltage, static and dynamic forward resistance, and reverse saturation current.
- To plot the forward- and reverse-bias V–I characteristics of a Zener diode and to determine its Zener breakdown voltage (V_Z) and Zener resistance (R_Z).
- To design, assemble, and test positive, negative, and biased clipper circuits, and positive and negative clamper circuits, and to record their input–output waveforms.

2. Apparatus / Components Required

S.No.	Component / Equipment	Specification	Quantity
1	PN junction diode	1N4007 (Si) or 1N4148	1
2	Zener diode	1N4733A ($V_Z = 5.1$ V, 1 W) or similar	1
3	Regulated DC power supply	0–30 V	1
4	Function generator	0–1 MHz, sine/square	1
5	Dual-trace CRO / DSO	20 MHz or better	1
6	Digital multimeter	—	2
7	Fixed resistors	1 k Ω , 4.7 k Ω , 10 k Ω	as needed
8	Breadboard, connecting wires, probes	—	as needed

3. Theory

3.1 PN Junction Diode

A PN junction diode conducts predominantly in one direction. Under forward bias, majority-carrier injection across the junction causes the current to rise exponentially once the applied voltage exceeds the cut-in (or knee) voltage V_γ — about 0.7V for silicon and 0.3V for germanium — after which current increases very rapidly for a small increase in voltage. Under reverse bias, only a small reverse

saturation current I_0 (due to minority carriers) flows, and this remains almost constant with increasing reverse voltage until the avalanche/Zener breakdown region is reached, beyond which the current rises sharply and can damage the device if not current-limited.

3.2 Zener Diode

A Zener diode is a heavily doped silicon PN junction specifically designed to operate in the reverse breakdown region without damage. Its forward characteristic is identical to that of an ordinary diode. In reverse bias, once the applied voltage reaches the Zener voltage V_z , the current increases sharply while the voltage across the diode remains nearly constant over a wide current range — this near-vertical knee is what makes the Zener diode useful as a voltage regulator/reference. Breakdown below about 5 V is dominated by the Zener effect (field emission across a narrow, heavily-doped junction); above about 6 V it is dominated by avalanche multiplication.

3.3 Clippers

A clipper (limiter) circuit removes or 'clips' the portion of a waveform that lies above and/or below a chosen reference level, using a diode in series or shunt with a load resistor, optionally biased with a reference (battery/Zener/resistor-divider). Series clippers pass the signal only when the diode conducts; shunt clippers divert the unwanted portion to ground/reference through the diode. Clippers are used for waveform shaping, protection of downstream circuits from over-voltage, and in transmitter/receiver limiting stages.

3.4 Clampers

A clamper (DC restorer) circuit shifts an AC waveform vertically by a fixed amount, without altering its shape, by adding a DC level derived from a diode–capacitor combination. During the half-cycle in which the diode conducts, the capacitor charges to (nearly) the peak input voltage; this stored charge then adds to (or subtracts from) the input during the rest of the cycle, restoring a DC reference to the signal. Positive clampers shift the waveform upward (fixing the minimum near a chosen level); negative clampers shift it downward (fixing the maximum near a chosen level).

Expected Input-Output Waveforms: Clippers and Clamper

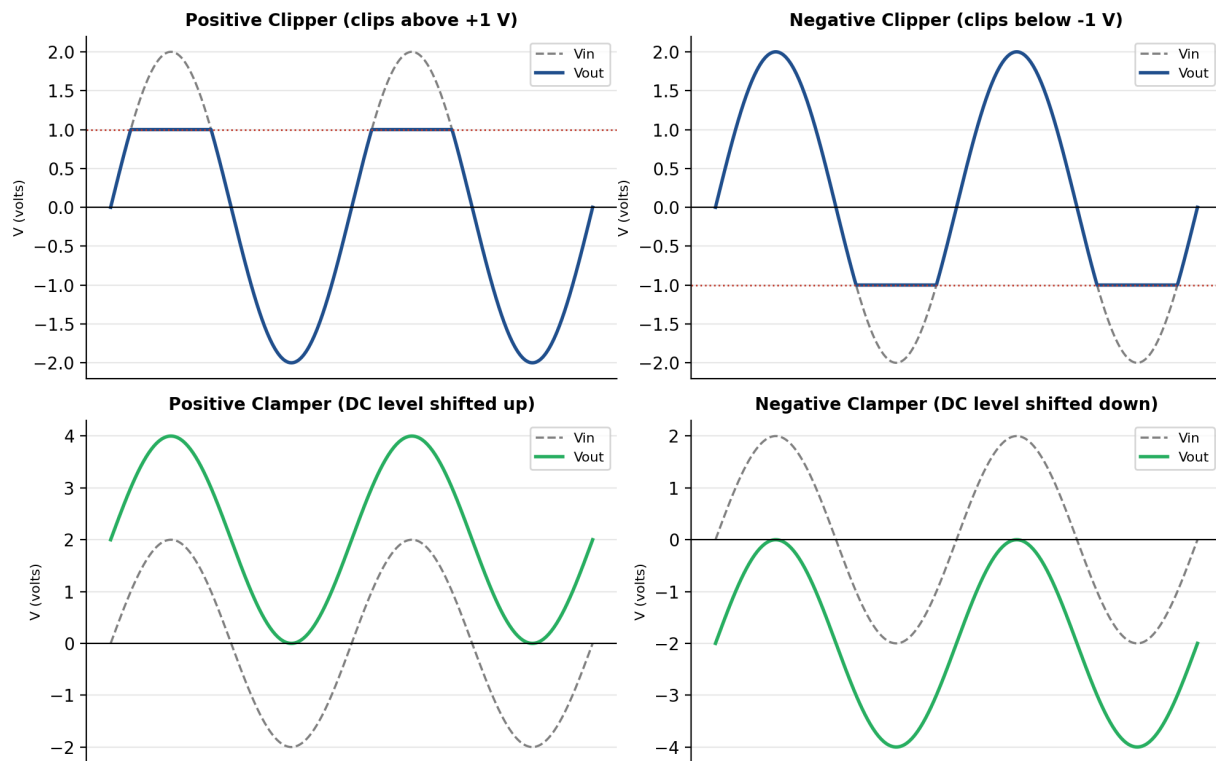


Fig. 3: Expected input (dashed) and output (solid) waveforms for representative clipper and clamper circuits.

4. Part A — PN Junction Diode: Forward Bias

Circuit Connections

1. Connect the diode in series with a current-limiting resistor $R = 1\text{ k}\Omega$ and a milliammeter (0–30 mA range), with the diode anode toward the positive supply terminal.
2. Connect a voltmeter (0–1 V range) directly across the diode terminals (not across the resistor).
3. Keep the DC supply output at 0 V and confirm all connections before switching on.

Procedure

4. Switch on the DC power supply with the output set to 0 V.
5. Increase the supply voltage in steps of 0.1 V from 0 V to 0.5 V. At each step, record the voltmeter reading (V_f) and milliammeter reading (I_f).
6. From 0.5 V to 0.75 V, reduce the step size to 0.05 V, since current changes rapidly in this region.
7. Beyond 0.75 V, adjust the supply so that the current increases in steps of approximately 2 mA per reading, continuing up to a maximum forward current of 20 mA. Do not exceed the diode's rated forward current.

8. Record at least 8–10 (V, I) pairs. Switch off the supply and tabulate the readings.
9. Plot I_f (Y-axis) versus V_f (X-axis) on graph paper to obtain the forward characteristic.

Observation Table — Forward Bias (PN Diode)

S.No.	Diode Voltage V_f (V)	Diode Current I_f (mA)
1		
2		
3		
4		
5		
6		
7		
8		

5. Part B — PN Junction Diode: Reverse Bias

Circuit Connections

10. Reverse the diode polarity from Part A: connect the diode cathode toward the positive supply terminal and the anode toward the negative terminal.
11. Connect a microammeter (0–100 μA range) in series with the diode, and a voltmeter (0–30 V range) directly across the diode.

Procedure

12. With the supply at 0 V, switch on the DC power supply.
13. Increase the reverse voltage in steps of 5 V from 0 V up to 30 V (or the diode's maximum rated reverse voltage, whichever is lower). At each step, record V_r and I_r .
14. If the current begins to rise sharply (breakdown), immediately reduce the voltage back toward 0 V to avoid damaging the diode; note the approximate breakdown voltage.
15. Switch off the supply and tabulate the readings.
16. Plot I_r (Y-axis) versus V_r (X-axis) on the same graph sheet as Part A, using an expanded/separate scale for the reverse-current axis if needed.

Observation Table — Reverse Bias (PN Diode)

S.No.	Diode Voltage V_r (V)	Diode Current I_r (μA)
1		
2		
3		
4		
5		
6		
7		
8		

6. Part C — Zener Diode: Forward and Reverse Bias**Forward Bias — Procedure**

17. Replace the PN diode with the Zener diode, keeping the same circuit as Part A (anode toward positive terminal, $R = 1\text{ k}\Omega$, milliammeter in series, voltmeter across the diode).
18. Repeat steps 1–6 of Part A exactly, recording 8–10 (V_f , I_f) pairs.

Reverse Bias — Procedure

19. Reconnect the Zener with its cathode toward the positive supply terminal, in series with $R = 1\text{ k}\Omega$ and a milliammeter (0–30 mA range); connect a voltmeter (0–10 V) directly across the diode.
20. Switch on the supply with the output at 0 V.
21. Increase the reverse voltage in steps of 0.5 V from 0 V up to 4 V, recording V_r and I_r at each step.
22. From 4 V to 5.5 V, reduce the step size to 0.1 V to accurately trace the sharp breakdown knee.
23. Beyond breakdown, increase the current in steps of about 2 mA up to a maximum of 20 mA, recording the corresponding (nearly constant) voltage each time.
24. Switch off the supply and tabulate all readings.
25. Plot the complete forward and reverse characteristic on one graph, and determine V_z and $R_z = \Delta V_z / \Delta I_z$ from the near-vertical breakdown region.
- 26.

Observation Table — Zener Diode (Forward and Reverse)

S.No.	V _f (V)	I _f (mA)	V _r (V)	I _r (mA)
1				
2				
3				
4				
5				
6				
7				
8				

7. Part D — Clipper Circuits**D1. Positive Clipper (clip above +1 V)**

27. Connect the function generator output in series with a resistor $R = 1\text{ k}\Omega$ to the output node.
28. From the output node, connect the diode cathode to a +1 V DC reference and the diode anode to the output node, so the diode conducts (clamping the output to +1 V) whenever the input exceeds +1 V.
29. Connect a bleed/load resistor $R_L = 10\text{ k}\Omega$ from the output node to ground.
30. Set the function generator to a 1 kHz sine wave, 4 V_{pp}, 0 V DC offset, and switch it on.
31. Connect CRO Channel 1 to the input node and Channel 2 to the output node, with both probe grounds at circuit ground.
32. Switch on the CRO and set both channels to the same Volts/div; overlay or display the two traces.
33. Record/sketch the input and output waveforms to scale and note the measured clipping level.

D2. Negative Clipper (clip below –1 V)

34. Reverse the diode direction from D1 (anode to the –1 V DC reference, cathode to the output node) so the output is clamped whenever the input falls below –1 V.
35. Keep the same R , R_L , and function generator settings as D1.
36. Repeat steps 5–7 of D1 and record the output waveform.

D3. Biased (Two-Level) Clipper

37. Combine the D1 and D2 circuits at the same output node: one diode referenced to +1 V (clips the top), the other referenced to -1 V (clips the bottom).
38. Repeat steps 5–7 of D1, and record the doubly-clipped output waveform, noting both clipping levels.

Observation Table — Clippers

Circuit	V _{in} (V _{pp})	Reference Level(s)	V _{out} (measured clip level)	Waveform sketch/attach
Positive clipper				
Negative clipper				
Biased clipper				

8. Part E — Clamper Circuits

E1. Positive Clamper

39. Connect a capacitor $C = 0.1 \mu\text{F}$ in series between the function generator output and the output node.
40. From the output node, connect the diode cathode to ground and anode to the output node (diode conducts on the negative half-cycle, charging C, and the output is shifted upward).
41. Connect a bleed resistor $R_L = 10 \text{ k}\Omega$ from the output node to ground.
42. Set the function generator to a 1 kHz sine wave, 4 V_{pp}, 0 V DC offset, and switch it on.
43. Connect CRO Channel 1 to the input node and Channel 2 to the output node; switch on the CRO and display both traces on the same voltage scale.
44. Record/sketch the input and output waveforms and measure the DC shift (difference between the input's and output's minimum levels).

E2. Negative Clamper

45. Reverse the diode direction from E1 (anode to ground, cathode to the output node), so the diode conducts on the positive half-cycle and the output is shifted downward.
46. Keep the same C, R_L , and function generator settings as E1.
47. Repeat steps 5–6 of E1 and record the output waveform, measuring the DC shift at the output's maximum level.

Observation Table — Clampers

Circuit	V _{in} (V _{pp})	Expected DC shift	Measured DC shift	Waveform sketch/attach
Positive clamper				
Negative clamper				

9. Calculations / Analysis

- From the PN diode forward data: read off the cut-in voltage V_γ ; compute static resistance (V/I) and dynamic resistance ($\Delta V/\Delta I$ near the operating point) at two or more points.
- From the PN diode reverse data: note the (near-constant) reverse saturation current and the reverse breakdown voltage, if observed.
- From the Zener data: locate the breakdown knee and read V_z ; compute the Zener resistance $R_z = \Delta V_z/\Delta I_z$ over the linear breakdown region.
- For each clipper, compare the measured clipping level to the value predicted from the bias reference and the diode's cut-in voltage ($V_{ref} \pm V_\gamma$).
- For each clamper, compare the measured DC shift to the theoretical shift ($\approx$ peak input amplitude, minus the diode drop).